The empirical recurrence for OEIS A230815, recovered from its tail

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Abstract

OEIS A230815 records "Empirical recurrence of order 95" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 98$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 97$. Its last coefficient is $c_{95} = -8216 \neq 0$, so no further tail satisfies anything shorter; any order-95 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A230815 is "Number of nX4 0..2 arrays x(i,j) with each element horizontally or vertically next to at least one element with value (x(i,j)+1) mod 3 and at least one element with value (x(i,j)-1) mod 3, and upper left element zero.". It has offset 1 and begins

$$0, 0, 12, 174, 816, 12210, 108006, 1202184, \ \dots$$

The entry states:

> Empirical recurrence of order 95 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 06:02:41, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 98$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 98$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 95: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 95.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 196$ exact terms from $n = 2$ on, it returns order 95 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{95} c_i \, a(n-i),$$

c_{1}	18	c_{25}	5942091607161	c_{49}	1051883821184161	c_{73}	−67584663446413
c_{2}	−79	c_{26}	−4947844528854	c_{50}	−1796553794058522	c_{74}	53446286764308
c_{3}	−176	c_{27}	1142756825828	c_{51}	2177605316472777	c_{75}	−42257659924963
c_{4}	2850	c_{28}	6387542989466	c_{52}	−1499469180113092	c_{76}	28756509233228
c_{5}	−15494	c_{29}	−16180475084523	c_{53}	−179081714933496	c_{77}	−16366933952314
c_{6}	53347	c_{30}	20048034084627	c_{54}	1878605261274138	c_{78}	7762547036040
c_{7}	−55317	c_{31}	−7240261177769	c_{55}	−2368452355742134	c_{79}	−2456370217794
c_{8}	−502057	c_{32}	−11476477229228	c_{56}	1230951503709545	c_{80}	85227557409
c_{9}	3640894	c_{33}	−6825061009201	c_{57}	622185699944060	c_{81}	596382358173
c_{10}	−15858253	c_{34}	84179094821605	c_{58}	−1999803414195785	c_{82}	−529701532419
c_{11}	60836322	c_{35}	−144155985315853	c_{59}	2168105944522770	c_{83}	210385389376
c_{12}	−215829761	c_{36}	60328903033897	c_{60}	−1002934609659740	c_{84}	−3712829218
c_{13}	619613373	c_{37}	168843289183034	c_{61}	−616661033381283	c_{85}	−35311738767
c_{14}	−1154299307	c_{38}	−381479222952212	c_{62}	1679399139587068	c_{86}	14064812869
c_{15}	−389762301	c_{39}	457750436336650	c_{63}	−1720578733465976	c_{87}	−1332786345
c_{16}	13486533430	c_{40}	−477026817645625	c_{64}	937655935719593	c_{88}	−875308528
c_{17}	−60102118879	c_{41}	503884896481127	c_{65}	−35212492245023	c_{89}	418851844
c_{18}	165168883690	c_{42}	−313453003384316	c_{66}	−497326142015237	c_{90}	−84657066
c_{19}	−307424093900	c_{43}	−380452575065900	c_{67}	629820629520043	c_{91}	251211
c_{20}	333214785485	c_{44}	1356988487552493	c_{68}	−497587235518168	c_{92}	3611050
c_{21}	76986824670	c_{45}	−1855521436143496	c_{69}	264723602832512	c_{93}	−847642
c_{22}	−1243300829057	c_{46}	1466591583734288	c_{70}	−70829990490635	c_{94}	111888
c_{23}	3107416719181	c_{47}	−522271457095801	c_{71}	−40808981967683	c_{95}	−8216
c_{24}	−5015411217865	c_{48}	−364059638254015	c_{72}	76390272944126		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 97$, and it is the recurrence OEIS A230815 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{95} - c_1 x^{95-1} - \cdots - c_{95}$. Its constant term is $-c_{95} = 8216$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 95 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 95, so $\deg q = 95$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 95, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 95, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -8216.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-95 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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